

Amal Nath VJ

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Profile Summary

Passionate **AI Developer** skilled in **Python, Machine Learning, and LLM-based systems**. Experienced in building intelligent applications using **LangChain, FastAPI, and Flask**, with hands-on exposure to **RAG pipelines and data automation**. Eager to apply agentic AI concepts to solve real-world challenges in collaborative, fast-paced environments.

Education

College of Engineering Trivandrum

2021 - 2025

B.Tech in Computer Science and Engineering

Experience

Siemens Technology and Services Pvt. Ltd | Intern

May 2024 ,Chennai

- Worked on smart building projects involving access controllers (Sipass), SDT (System Design Tool), and surveillance systems.
- Designed and implemented an AuditHistory table to track user actions along with timestamps and user info.

Projects

Coder Buddy – AI-Powered Development Assistant 

Python, LangChain, LangGraph

- Built an AI-driven coding assistant that scaffolds complete software projects and provides context-aware code generation using Python, LangGraph, and LangChain.
- Engineered a modular agent framework integrating prompt templates, workflow orchestration, state management, and tool-based execution for reproducible development workflows.

AskMyDocs – AI-Driven Answering Platform 

Python, LangChain, ChromaDB, RAG

- Developed and implemented an AI-powered PDF Question Answering system leveraging Retrieval-Augmented Generation (RAG) with LangChain, ChromaDB, and GROQ/OpenAI LLMs.
- Automated document ingestion, semantic chunking, and vector-based search to deliver accurate, context-aware responses with verified source citations.

Fake Job Posting Detection System - Machine Learning 

Python, Scikit-learn, NLP, Flask

- Built an end-to-end ML system to detect fraudulent job postings using TF-IDF, categorical encoding, and scikit-learn classifiers, achieving high detection accuracy.
- Deployed a Flask web app with RESTful API for real-time predictions, integrating preprocessing, model inference, and user-friendly frontend.

Log Classification with Hybrid Framework 

Python, FastAPI, NLP, LLM

- Built an intelligent log classification system to categorize system logs (e.g., errors, warnings, notifications).
- Designed a 3-tier hybrid architecture combining Regex for structured logs, Sentence-Transformer + Logistic Regression for semantic patterns, and LLM (DeepSeek-R1-Distill-LLaMA-70B) for poorly-labeled legacy data with confidence-based fallback.
- Deployed the system using FastAPI for real-time log classification.

GroceryFlow – Grocery Store Management System 

Python, Flask, MySQL, JavaScript

- Developed a full-stack web application with a RESTful Flask backend and responsive frontend to manage inventory, product catalog, and order processing.
- Designed a normalized MySQL database and implemented a clean DAO pattern with robust CRUD APIs, error handling, and CORS support.
- Deployed the application with modular, maintainable code architecture, achieving seamless frontend-backend integration and reliable operations.

Technical Skills

Languages: Python, SQL, JavaScript, HTML/CSS
Frameworks/Libraries: Flask, FastAPI, Sklearn, spaCy, Langchain, Huggingface
Developer Tools: GitHub, VS Code, Jupyter Notebook, Google Colab
Domains: Machine Learning, Natural Language Processing , Large Language Model , Gen AI

Achievements & Leadership

- Siemens Scholar

2021–2025

- Selected among top engineering students in India based on academic merit and socio-economic background.
- Leader, CET Official Dance Crew

2024–2025

- Led the team to finals of India’s top dance competition, Destiny Dance Competition.
- Coordinator, HOPE (Alumni Charitable Organization)

2023–2025

- Coordinated scholarship programs supporting financially challenged students, focusing on technical and analytical merit.